

School of Information Technology and Engineering (SITE) Model Exam 2

Object-oriented programming

1. Consider the following Java program and then choose the correct statement from the list that follows:

```
public class Hello
{
    public Static void main(String[] args)
    {
        System.out.print("Hello ");
        System.out.print("world")

    }
}
```

- A. this program contains more than one syntax error;
- B. this program contains one and only one syntax error;
- C. this program compiles successfully and produces the following output:
Hello world
- D. this program compiles successfully and produces the following output:
Hello
world
- E. none of the above.

2. Consider the following Java program and then choose the correct statement from the list that follows:

```
public class AnyProg
{
    public static void main (String[] args)
    {
        int x, y;
        x = 7;
        x++;
        y = x;
        x = x * 3;
        System.out.println ("the value of y is " + y);

    }
}
```

- A. this program compiles successfully and produces the following output:
the value of y is 24
- B. this program contains at least one syntax error ;
- C. this program compiles successfully and produces the following output:
the value of y is 8
- D. this program compiles successfully and produces the following output:
the value of y is + y
- E. this program compiles successfully and produces the following output:
the value of y is 7

3. Which of the following statements is NOT correct?

- A. the object-oriented approach does not separate the behaviour of a system from the data;
- B. Java is an object-oriented programming language;
- C. C++ is an object-oriented programming language;
- D. Java is the only object-oriented programming language;
- E. the object-oriented approach makes a system easier to maintain.

The following UML design of an Employee class is referred to in questions 4 and 5. You can assume that when this class is implemented, the attributes will be declared as private and the methods as public. You can also assume that the get- methods simply return the value of the relevant attribute, and that the calculateAge method returns the input value minus the value of the yearOfBirth attribute.

<i>Employee</i>
<i>name : String</i> <i>address : String</i> <i>phone : String</i> <i>yearOfBirth : int</i>
<i>Employee(String, String, String, int)</i> <i>setAddress(String)</i> <i>getName() : String</i>

4. Assuming that the Employee class shown above is accessible to the Java compiler, which of the following statements would NOT result in a compiler error?
- A. Employee emp1 = new Employee("845","5034","0101",0);
 - B. Employee emp1 = new Employee("Patel","Essex","02085332455","1970");
 - C. Employee emp1 = new Employee("Patel","Essex","1970");
 - D. emp1 Employee = new Employee("Patel","Essex","02085332455",1970);
 - E. Employee emp1 = new Employee("Patel",Essex,02085332455,1970);
5. Assuming that an object called emp1, of class Employee, has already been created, which of the following statements would cause the address of emp1 to be displayed on the screen?
- A. System.out.print(address);
 - B. System.out.print(emp1.Address);
 - C. System.out.print(emp1.getAddress());
 - D. System.out.print(emp1.getAddress);
 - E. System.out.print(getAddress());

The following definition of a class called SomeClass is referred to in questions 6 and 7.

```
class SomeClass
{
    private int one;
    private int two;

    public SomeClass()
    {
        one = 2;
        two = 4;
    }

    public SomeClass(int x, int y)
    {
        one = x;
        two = y;
    }

    public void setOne(int x)
    {
        one = x;
    }

    public void setTwo(int x)
    {
        two = x;
    }
}
```

```

public void equalize()
{
    one = two;
}

public int calcSum()
{
    return (one + two);
}

public int calcDiff()
{
    return (two - one);
}

}

```

6. Consider the following program which uses SomeClass (as defined above), and choose the correct statement from those that follow:

```

public class SomeProg
{
    public static void main(String[] args)
    {
        SomeClass myClass = new SomeClass();
        myClass.equalize();
        System.out.print(myClass.calcSum());
    }
}

```

```
}  
}
```

- A. the program will not compile successfully;
 - B. the program compiles successfully and produces an output of 4;
 - C. the program compiles successfully and produces an output of 2;
 - D. the program compiles successfully and produces an output of 8;
 - E. the program compiles successfully and produces an output of 6.
7. Consider the following program which uses SomeClass (as defined above), and choose the correct statement from those that follow:

```
public class SomeProg  
{  
    public static void main(String[] args)  
    {  
        SomeClass myClass = new SomeClass(3);  
        myClass.setTwo(1);  
        System.out.print(myClass.calcDiff());  
    }  
}
```

- A. the program will not compile successfully;
- B. the program compiles successfully and produces an output of 2;
- C. the program compiles successfully and produces an output of -2;

- D. the program compiles successfully and produces an output of 1;
- E. the program compiles successfully and produces an output of -1.
8. Consider the following explicit creation of an array and then choose the correct statement from the list that follows

```
int[] someArray = { 2, 13, 9, 11, 10 };
```

- A. the value of someArray.length is 6;
- B. the value of someArray[2] is 13;
- C. an array cannot be created this way in Java;
- D. the value of someArray[11] is 4;
- E. none of the above.
9. Consider the following method, setValue, which adds a value into an array list

```
public void setValue (int position, String valueIn)
{
    list[position-1] = valueIn;
}
```

Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an ArrayIndexOutOfBoundsException exception:

- A. the return type being declared as **void**;
- B. the method having more than one parameter;
- C. the value of the first parameter ;
- D. the type of the parameter position being declared as an **int**;
- E. not declaring the method **static**;

10. Consider the program below, then answer the question that follows:

```
package test;
public class SomeProg
{
    public static void main (String[] args)
    {
        for (int i = 0; i < args.length; i++)
        {
            System.out.println(args[i]);
        }
    }
}
```

How would this program be run from the command line to produce the following output?

Java

in two semesters

- A. **java test.SomeProg Java in two semesters**
- B. **java SomeProg "Java in two semesters"**
- C. **java test.SomeProg Java, in two semesters**
- D. **java test.SomeProg "Java" "in two semesters"**
- E. none of the above.

Web Programming

1. What will be the output of the following code snippet?

```
<script type="text/javascript" language="javascript"> var a = "exercise"; var result =
    a.substring(3, 5); document.write(result);

</script>
```

- a. **erc**
- b. **rc**

- c. er
 - d. rcis
2. What will be the output of the following code snippet?
- ```
const set = new Set(); set.add(7); set.add('java');
set.add({ name: 'script' }); for (let item of set)
{ console.log(item + 5);
}
```
- a. 12 NaN NaN
  - b. 12 java5 [Object Object]5
  - c. 12 java5 NaN
  - d. 12 undefined undefined
3. By default Hyperlinks are displayed with an underline. How do you remove the underline from all hyperlinks by using CSS code ?
- a. a {text: no-underline;}
  - b. a{text-decoration:none;}
  - c. a {text-style: no-underline;}
  - d. a {text-decoration: no-underline;}
4. In HTML form the <input type="text"> is used for
- a. One line text
  - b. Block of text
  - c. One paragraph
  - d. None of the above
5. Which HTML tag is used to define an abbreviation or an acronym?
- a. <abbreviation>
  - b. <abbr>
  - c. <acronym>
  - d. <acr>
6. Which request URI component is optional?
- a. URI host
  - b. URI scheme
  - c. Query string
  - d. Resource path
7. Which component is used for specifying the protocol used for transmitting the request?
- a. URI host
  - b. URI Scheme
  - c. Resource path
  - d. GET

8. Which of the following explains Cookie's nature?
  - a. Non volatile
  - b. Volatile
  - c. Intransient
  - d. Transient
9. Within Oauth, what component validates the user's identity?
  - a. Client
  - b. Authorization Server
  - c. Resource Server
  - d. Browser
10. What is one benefit of GraphQL over the REST approach?
  - a. More stable APIs
  - b. Flexible querying/responses
  - c. Compatible with more gateways
  - d. More secure by default

#### Mobile Programming

1. APK stands for -
  - a. Android Phone Kit
  - b. Android Platform Kit
  - c. Android Page Kit
  - d. Android Package Kit
2. What is an activity in android?
  - a. android class used to configure the android application
  - b. android package file holding all the packages used
  - c. A single screen in an application with supporting java code
  - d. None of the above
3. On which of the following, developers can test the application, during developing the android applications?
  - a. Third-party emulators
  - b. Emulator included in Android SDK
  - c. Physical android phone
  - d. All of the above
4. Which one of the following can be used to persist data in Android device
  - a. Microsoft SQL Client
  - b. Shared Preference
  - c. MongoDB Database
  - d. Firebase/FireStore
5. How to pass the data between activities in Android?
  - a. Content Provider
  - b. Broadcast receiver
  - c. Intent
  - d. PostgreSQL Database
6. What is Manifest.xml in android?

- a. It has information about layout in an application
  - b. It has the information about activities in an application
  - c. It has all the information about an application
  - d. None of the above
7. In which directory XML layout files are stored?
- a. /assets
  - b. /src
  - c. /res/values
  - d. /res/layout
8. All layout classes are subclasses of which of the following?
- a. android.view.Widget
  - b. android.view.Layout
  - c. android.view.RelativeLayout
  - d. android.view.ViewGroup
9. Which of the following is the first callback method that is invoked by the system during an activity life-cycle?
- a. onClick() method
  - b. onCreate() method
  - c. onStart() method
  - d. onRestart() method
10. Which of the following android components displays the part of an activity on screen?
- a. View
  - b. Manifest
  - c. Intent
  - d. Fragment

#### Software and System Security

1. A security objective(s) that is(are) not controlled by cryptographic techniques
- a. Confidentiality
  - b. Integrity
  - c. Nonrepudiation
  - d. Availability
2. The overlap of threat and an asset's weakness is \_\_\_\_\_
- a. Vulnerability
  - b. Loss
  - c. Risk
  - d. Exploit
3. Which of the following is considered as the unsolicited commercial email?
- a. Virus

- b. Spam
  - c. Malware
  - d. All of the above
4. \_\_\_\_\_ is a security measure that verifies whether you have permission to access the specific resources
- a. Authentication
  - b. Authorization
  - c. Non-repudiation
  - d. Accountability
5. A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.
- a. Security Attack
  - b. Security Service
  - c. Threat
  - d. Security Mechanisms
6. DES has an initial and final permutation block and \_\_\_\_\_ rounds.
- a. 15
  - b. 14
  - c. 16
  - d. All
7. Which of the following Algorithms belong to symmetric encryption?
- a. RSA
  - b. DES
  - c. 3DES
  - d. All except a
8. In which way does the Combined Encryption combine symmetric and asymmetric encryption?
- a. The secret key is asymmetrically transmitted, the message itself symmetrically.
  - b. First, the message is encrypted with symmetric encryption and afterwards it is encrypted asymmetrically together with the key

- c. The secret key is symmetrically transmitted, the message itself asymmetrically.
  - d. First, the message is encrypted with asymmetric encryption and afterwards it is encrypted symmetrically together with the key.
9. Using public-key infrastructure (PKI) encryption, which key is used by the sender for authentication of the receiving party?
- a. Sender's private key
  - b. Recipient's private key
  - c. Sender's public key
  - d. Recipient's public key
10. In using public-key infrastructure (PKI) encryption, which key is not used by the recipient for Decrypting a message?
- a. Sender's private key
  - b. Recipient's private key
  - c. Sender's public key
  - d. Recipient's public key

#### Fundamentals of Software Engineering

1. Why do we need requirements?
  - a. to understand precisely what is required of the software
  - b. communicate this understanding precisely to all development parties
  - c. control production to ensure that system meets specs
  - d. all
2. \_\_\_\_\_ is creating a scaled-down or incomplete version of a system to demonstrate or test its aspects.
  - a. prototyping
  - b. mocking
  - c. configuration
  - d. staging
3. \_\_\_\_\_ Defined in a discipline method to get a program that have the quality of readability, maintainability and debug-ability.
  - a. coding
  - b. structured programming
  - c. logical programming
  - d. none
4. Which of the following is true about testing.

- a. Alpha testing continues until customer agrees that system implementation is as per his/her expectation.
  - b. Beta testing may be defined as system testing which is done by the customer on customer's own sites.
  - c. Alpha testing results in minor design changes.
  - d. all
- 5. Which one of the following is included in software implementation process?
  - a. training
  - b. designing
  - c. planning
  - d. requirement gathering
- 6. Which of the following tasks does not belong in requirement elicitation activities?
  - a. Ethnographic study
  - b. Survey
  - c. Interview
  - d. None of the above
- 7. Which one of the following is not shown in class diagram?
  - a. Multiplicity
  - b. Class instances
  - c. Relationship
  - d. Operations and attributes
- 8. What is the role of a usecase diagram?
  - a. It shows interactions among components of the software
  - b. It shows interactions among objects of the software
  - c. It shows workflow of events to accomplish a task in the software
  - d. None of the above
- 9. Which one of the following is not true about non-functional requirements?
  - a. Failing to meet non functional requirements may render the whole system unusable
  - b. A single non-functional requirement may affect the whole system
  - c. Non-functional requirements lists the features of the system
  - d. None of the above
- 10. Which one of the following is not a fundamental activities in Software Engineering?
  - a. Software Specification
  - b. Software Evolution
  - c. Software Design and Implementation
  - d. None of the above

## Computer Programming

1. Which of the following variables names are valid?

- a. \*name      b. \_123      c. \_money\$      d. Full-name
- e. try      f. none

2. What is the output of the following code snippet

```
x= 5 * 2 + 3 // 4 + 10 ** 2
print(x)
```

- a. 20      b. 30      c. 110.75      d. None

3. What is the output of the following code snippet?

```
x= int(float("10.0")) + float(int("20")) // 2
print(x)
```

- a. 20      b. 15      c. 15.0      d. 20.0      E. Error

4. What is the output of the following code snippet?

```
x= not 2 + 3 == 5 or 5 + 2 =< 7 and 7 + 8 == 15
print(x)
```

- e. True      f. False      g. Error      h. None

5. What is variable in python?

- a. A named location in the memory that can store data
- b. A piece of data stored in the memory
- c. A function that is used to store data in the memory
- d. A Statement that can be used to store data in the memory

6. Which keyword is used to define a function in python?

- a. function
- b. define
- c. procedure
- d. None of the above

7. What does the following code produce as output?

```
programming = ['Software', 'Electrical', 'Civil']
print (max(programming))
```

- a. Software
- b. Electrical
- c. Civil
- d. None of the above

8. What results from the next list assignment?

```
siTE = ['A', 'B', 'C', 'D']
siTE [1:4] = ['E', 'F', 'G']
print(siTE)
```

- a. ['A', 'E', 'F', 'G', 'B', 'C', 'D']
- b. ['A', 'E', 'F', 'G']
- c. ['B', 'C', 'D', 'E', 'F', 'G']
- d. None of the above

9. Pick the right result from the ensuing string operations.

```
yourString = "Ethioipa"
stringList = ["Kenya", "South Africa", "Ethioipa"]
print(stringList[1] == yourString)
print(stringList[1] is yourString)
```

- a. True False
- b. False False
- c. True True
- d. None of the above

10. Which of the following string operations will get the right result?

```
a= 'Welcome'
print(a*2)
```

- a. Welcome
- b. TypeError unsupported operand type(s)
- c. a\*2
- d. None of the above